

SW Engineering CSC648-848 Spring 2026

Gator Freighter

Team 14

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Milestone 2 Part I

Date Submitted	April 5, 2026
Date Revised	TBD

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1. Executive Summary:

College students will be able to buy, sell, and trade everyday items from textbooks to electronics to furniture and more, with Gator Freighter. This online application will ensure that students can rely on quick, affordable, and safe transactions. General platforms like Facebook Marketplace or Craigslist expose students to strangers, unreliable transactions, and inconvenient or unsafe meetup locations. At San Francisco State University, where the cost of living and education is already a significant burden, there is now a clear, hassle-free, campus-focused solution with Gator Freighter.

Gator Freighter is a web-based marketplace exclusively for the SFSU community. The platform makes transactions simple and intuitive: users browse listings, connect with sellers through a built-in messaging system, and coordinate exchanges at designated safe meetup locations. Every user is verified through their SFSU emails, meaning every person on the platform is a member of the university community. This design is also made to reduce the time and effort students spend searching for relevant items, as users are presented with items that reflect common student needs. This creates a layer of trust and accountability that no general-purpose marketplace can offer to students.

Beyond improving individual transactions, Gator Freighter will make it easier for students to support each other, strengthening the SFSU community. Instead of relying on outside platforms, students can keep resources within their own campus network, helping others save money while building trust and familiarity. This will create a more connected environment where students aren't just users of a marketplace, but active participants in a shared system that benefits the whole campus.

Gator Freighter is being developed by a student startup team from San Francisco State University's Computer Science department. As active members of the SFSU community, we understand firsthand the frustrations of overpaying for textbooks, struggling to furnish an apartment on a tight budget, or simply trying to find a buyer for last semester's lab kit. We are building the practical solution we wish existed when we arrived on campus. We believe Gator Freighter will create lasting value to SFSU's community and for those who come after us.

2. List of main data items and entities:

Users - Represents registered SFSU users.

- id (required): unique identifier
- first_name (required): user's first name
- last_name (required): user's last name
- school_email (required): SFSU email used for verification
- password_hash (required): encrypted password
- profile_image_url (optional): link to profile image
- is_student_verified (required): indicates verified SFSU status
- account_status (required): account state (active or suspended)
- is_admin (required): indicates admin privileges
- storefront_enabled (optional): whether storefront is enabled
- storefront_name (optional): name of storefront
- storefront_description (optional): description of storefront
- storefront_banner_url (optional): banner image
- storefront_logo_url (optional): logo image
- created_at (required): timestamp of creation
- updated_at (required): timestamp of last update

Items (Listings) - Represents listings created by users.

- id (required): unique identifier
- seller_id (required): reference to user who created listing
- title (required): listing title
- description (optional): description of the item
- price (optional): cost of the item
- category_id (required): category reference
- course_id (optional): associated course
- listing_type (required): type of listing (sale, trade, or both)
- pickup_location_id (optional): meetup location
- status (required): active, sold, or removed
- is_featured (optional): featured listing flag
- created_at (required): creation timestamp
- updated_at (required): last update timestamp

Categories- Organizes listings.

- id (required): unique identifier
- name (required): category name

Courses - Represents SFSU courses.

- id (required): unique identifier
- course_code (required): course identifier (e.g., CSC648)
- course_name (required): full course name

Pickup Locations - Represents campus meetup points.

- id (required): unique identifier
- name (required): location name
- location (required): description of location
- description (optional): additional details
- is_active (required): indicates availability

Listing Images - Stores images for listings.

- id (required): unique identifier
- item_id (required): associated listing
- image_url (required): file path to image
- sort_order (optional): display order

Conversations - Represents communication channels between users.

- id (required): unique identifier
- item_id (required): associated listing
- buyer_id (required): buyer reference
- seller_id (required): seller reference
- last_message_at (optional): timestamp of last message
- created_at (required): creation timestamp

Messages - Stores messages between users.

- id (required): unique identifier
- conversation_id (required): conversation reference
- sender_id (required): sender of message
- body (required): message content
- is_read (required): read status
- created_at (required): timestamp

Trade Requests - Handles item exchanges.

- id (required): unique identifier
- requester_id (required): requesting user
- requested_item_id (required): desired item
- offered_item_id (required): offered item
- status (required): pending, accepted, declined, or cancelled
- created_at (required): timestamp
- updated_at (required): timestamp

Reports - Stores moderation reports.

- id (required): unique identifier
- reporter_id (required): reporting user
- reported_user_id (optional): reported user
- reported_item_id (optional): reported item
- reason (required): description of issue
- status (required): report state
- created_at (required): timestamp

3. Functional Requirements - prioritized:

Unregistered users:

1. Unregistered users shall be able to browse available listings.
2. Unregistered users shall be able to search listings by keyword.
3. Unregistered users shall be able to filter listings by category.
4. Unregistered users shall be prompted to register or log in when attempting restricted actions.
5. Unregistered users shall be prompted to verify they are an SFSU student using student email

Registered Users:

6. Registered users shall be able to create listings
7. Registered users shall be able to reply to listings
8. Registered users shall be able to create trade requests
9. Registered users shall be able to view their dashboard/account
10. Registered users shall be able to update their listings price.
11. Registered users shall be able to categorize listings under club and/or major categories.
12. Registered users shall be able to create/manage storefront
13. Registered users shall be able to view message threads
14. Registered users shall be able to search listings by keyword.
15. Registered users shall be able to filter listings by category.
16. Registered users shall be able to send messages to other users via listing.
17. Registered users shall be able to delete listings.
18. Registered users shall be able to filter listings by course number.
19. Registered users shall be able to report inappropriate listings or users.
20. Registered users shall be able to specify whether a listing is for sale or for trade.
21. Registered users shall be able to accept or decline trade proposals.
22. Registered users shall be able to select an available campus pickup location for meetups.

UPDATE:

Priority 1 (MUST HAVE)

Unregistered Users

- Users shall be able to browse listings
- Users shall be able to search listings by keyword
- Users shall be able to filter listings by category
- Users shall be able to view listing details

Registered Users

- Users shall be able to register using SFSU email

- Users shall be able to create listings
- Users shall be able to view listing details
- Users shall be able to send messages to sellers
- Users shall be able to select campus meetup locations
- Users shall be able to manage listings

Priority 2 (DESIRED)

Registered Users

- Users shall be able to create trade requests
- Users shall be able to filter listings by course

Priority 3 (OPTIONAL)

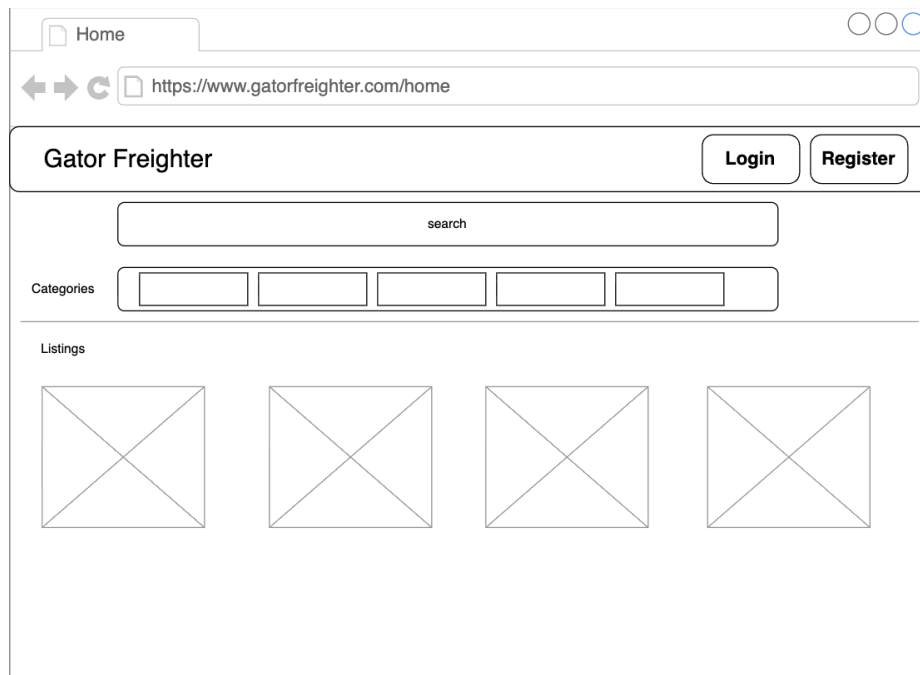
Registered Users

- Users shall be able to create storefronts
- Users shall be able to report users/listings

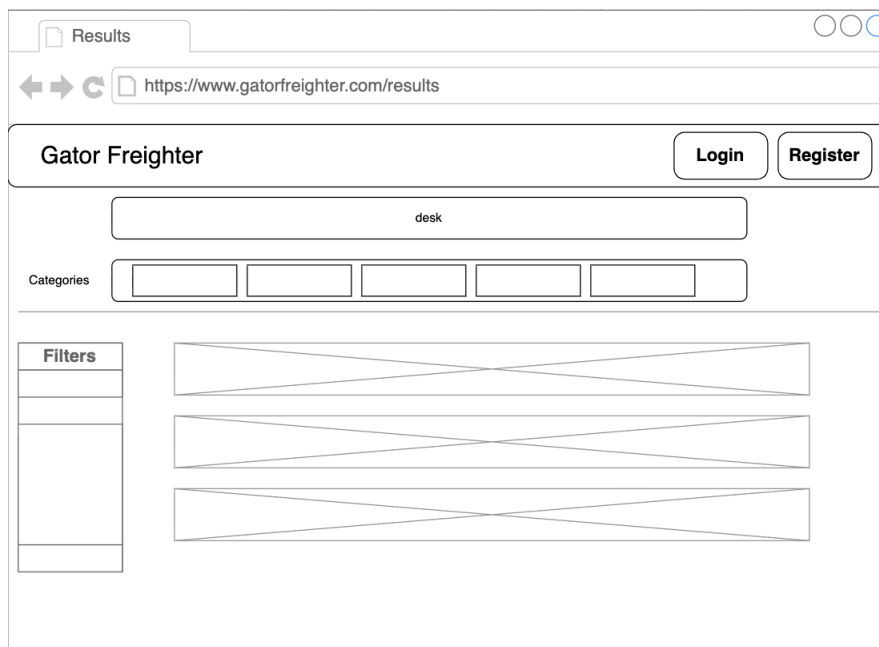
4. UI Storyboards:

Storyboard 1: Browse/Search listings - This storyboard illustrates how a user searches for items, filters results, and views item details before deciding to contact a seller.

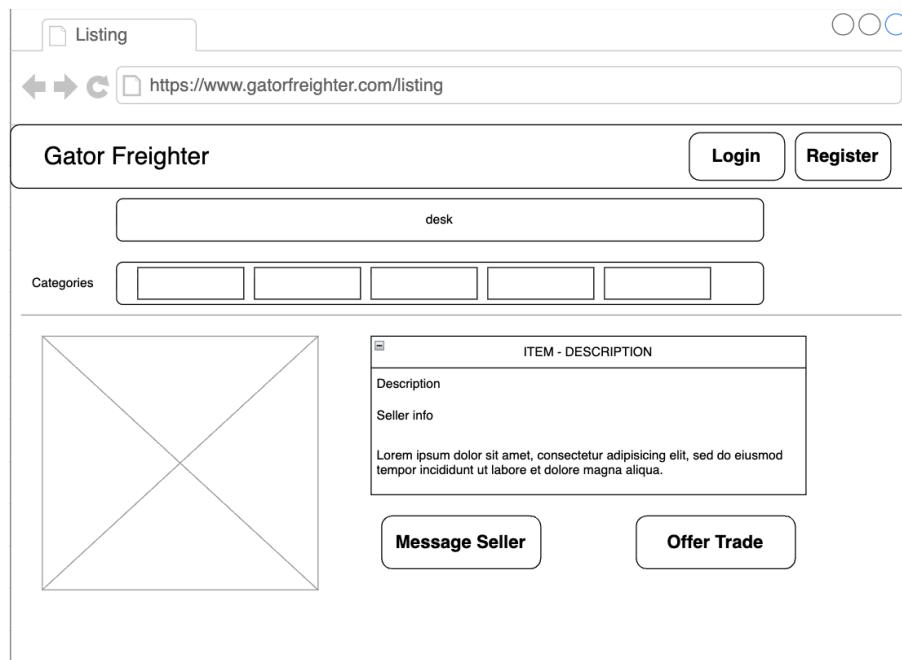
User lands on the home page and enters a search query. Users can filter listings using SFSU-specific tags.



User views a list of matching items with basic details: title, price, image.

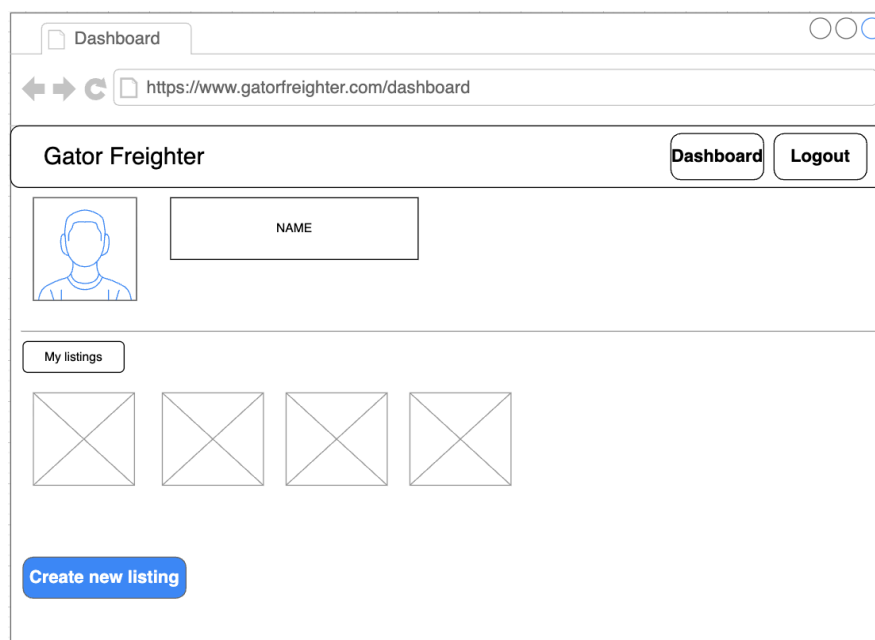


User selects a listing to view full details including description, price, and seller information. The user is given options to message the seller or initiate trade.



Storyboard 2: Create Listing - Illustrates how a user creates a new listing and manages it through their dashboard/storefront.

The user navigates from the home page to select create listing. The user must be logged in or is prompted to log in/register.



The user enters listing details such as title, description, price, and category (can add SFSU-specific tags like course codes).

The screenshot shows a web browser window with the address bar displaying `https://www.gatorfreighter.com/newlisting`. The page header includes the "Gator Freightier" logo and "Dashboard" and "Logout" buttons. The main content area features a "Create New Listing" form with the following fields: "Title", "Price", "Category", and "Image" (with a "[x]" icon and an "[upload]" button). The "Desc." field contains placeholder text: "Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum." At the bottom of the form are "SUBMIT" and "CANCEL" buttons.

After submitting the listing, the user is redirected to their dashboard/storefront where the new listing appears.

The screenshot shows the user's dashboard in the Gator Freightier application. The browser address bar displays `https://www.gatorfreighter.com/dashboard`. The header includes the "Gator Freightier" logo and "Dashboard" and "Logout" buttons. The main content area features a user profile section with a placeholder image and a "NAME" field. Below this is a "My listings" section with five placeholder images for listings. The fifth placeholder is labeled "NEW LISTING*". At the bottom of the dashboard is a blue "Create new listing" button.

Storyboard 3: Buyer - This storyboard shows how a user views item details, is prompted to register or login, and initiates a conversation with a seller through messages.

The user views item details, then selects “message seller” to initiate interaction, and is then prompted to register or login before accessing messaging features.

The screenshot shows a web browser window with the title 'Register' and the URL 'https://www.gatorfreighter.com/register'. The page header includes the 'Gator Freightier' logo and 'Login' and 'Register' buttons. The main content area features a 'Register Form' with the following elements:

- A heading 'Register Form'.
- A label 'Enter a valid SFSU email' above a text input field containing 'tbrown@sfsu.edu'.
- A label 'Password:' above a password input field with masked characters '*****'.
- A label 'New User' above a blue 'SIGN UP' button.

The user completes authentication using an SFSU email address and is redirected back to item details.

The screenshot shows a web browser window with the title 'Listing' and the URL 'https://www.gatorfreighter.com/listing'. The page header includes the 'Gator Freightier' logo and 'Dashboard' and 'Logout' buttons. The main content area includes:

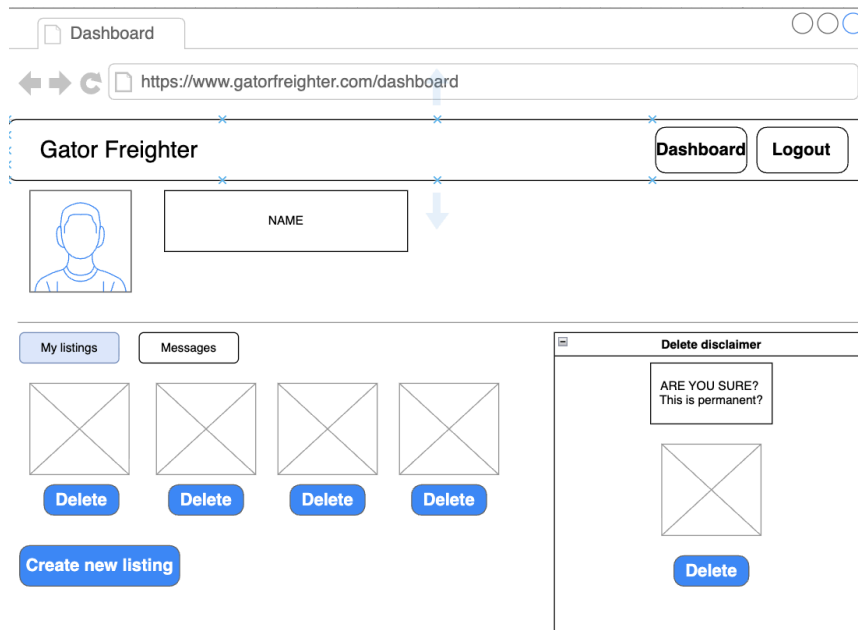
- A search bar with the placeholder text 'desk'.
- A 'Categories' section with five empty rectangular buttons.
- A large square placeholder with a diagonal 'X' on the left side.
- An 'ITEM - DESCRIPTION' box containing:
 - A 'Description' label.
 - A 'Seller info' label.
 - Placeholder text: 'Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.'
- Two buttons below the item description: 'Message Seller' and 'Offer Trade'.

The user selects “message seller” and is redirected to Chat where they can send and receive messages.

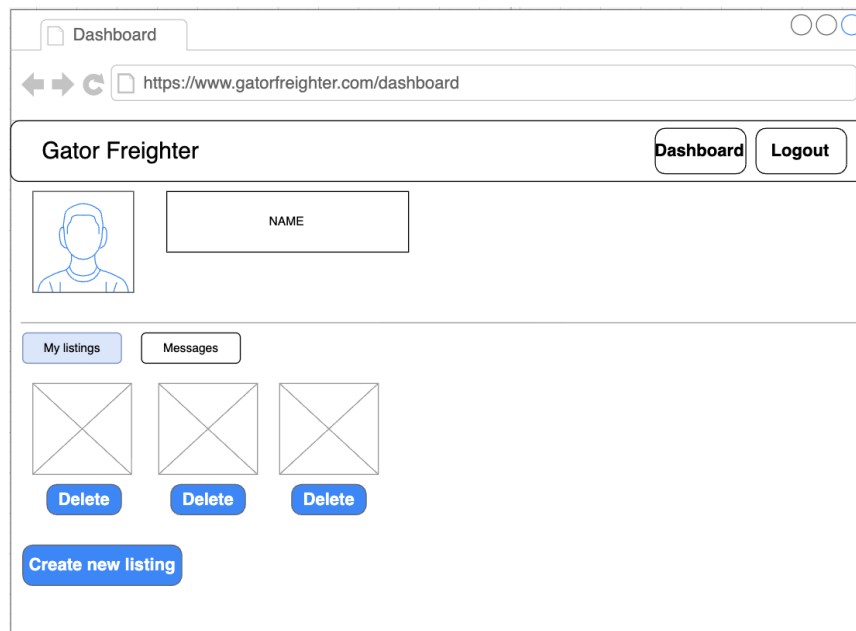


Storyboard 4: Seller responding to Buyers - This storyboard displays a user accessing their dashboard to optionally update their listings, then navigates to their inbox to respond to potential buyers.

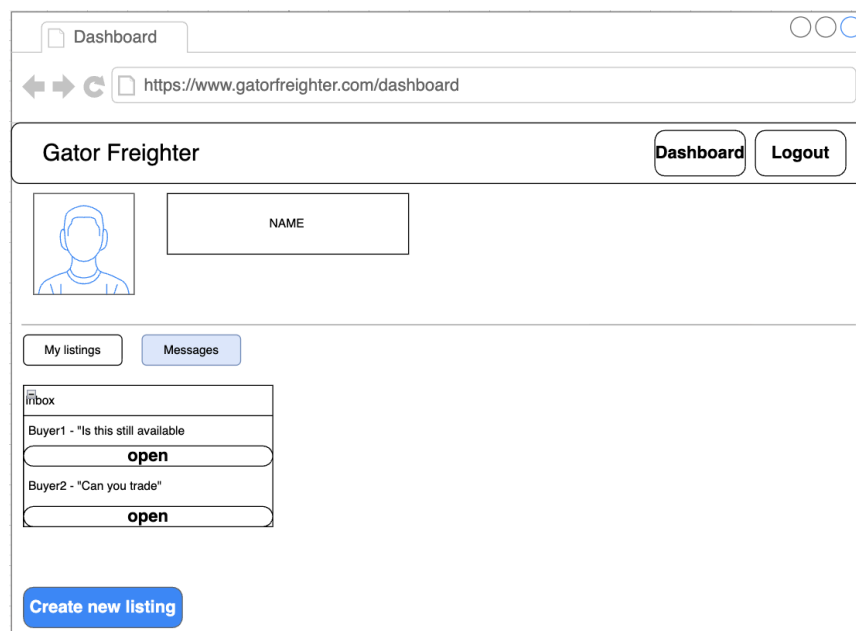
The user accesses the dashboard to delete a listing.



The user sees the listing they deleted is gone from their dashboard.



After the update, the user goes to check their messages.



Storyboard 5: Trade - This storyboard illustrates how a user navigates from item details to initiate a trade proposal instead of choosing to message the seller.

The user enters a search query for their desired item.

This screenshot shows the 'Results' page of the Gator Freightier website. The browser's address bar displays 'https://www.gatorfreighter.com/results'. The page header includes the site name 'Gator Freightier' and navigation links for 'Dashboard' and 'Logout'. Below the header is a search bar containing the text 'desk'. Underneath the search bar is a 'Categories' section with five empty selection boxes. On the left side, there is a 'Filters' sidebar with several empty filter boxes. The main content area displays three placeholder images, each represented by a rectangle with an 'X' inside, indicating where search results will be shown.

The user then clicks on a listing, entering item details.

This screenshot shows the 'Listing' page of the Gator Freightier website. The browser's address bar displays 'https://www.gatorfreighter.com/listing'. The page header includes the site name 'Gator Freightier' and navigation links for 'Login' and 'Register'. Below the header is a search bar containing the text 'desk'. Underneath the search bar is a 'Categories' section with five empty selection boxes. The main content area is divided into two columns. The left column features a large placeholder image, represented by a rectangle with an 'X' inside. The right column contains a section titled 'ITEM - DESCRIPTION' with fields for 'Description' and 'Seller info'. Below these fields is a block of placeholder text: 'Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.' At the bottom of the right column are two buttons: 'Message Seller' and 'Offer Trade'.

The user then starts a trade message as a trader.



5. High level Architecture, Database Organization summary only:

DB organization:

1. Users

Stores information about registered SFSU users and storefront details

- id
- first_name
- last_name
- school_email
- password_hash
- profile_image_url
- is_student_verified
- account_status
- is_admin
- created_at
- updated_at

2. storefront

- storefront_enabled
- storefront_name
- storefront_description
- storefront_banner_url
- storefront_logo_url

3. categories

Defines item categories used for browsing and filtering

- id
- name

4. pickup_locations

Designated meetup locations

- id
- name
- location
- description
- is_active

5. courses

Stores SFSU courses associated with academic-related listings

- id
- course_code
- course_name

6. items

Represents listings created by users

- id
- seller_id

- title
- description
- price
- category_id
- course_id
- listing_type
- pickup_location_id
- status
- is_featured
- created_at
- updated_at

7. listing_images

Stores image paths associated with listings

- id
- item_id
- image_url
- sort_order

8. conversations

Represents communication channels between buyers and sellers

- id
- item_id
- buyer_id
- seller_id
- last_messages_at
- created_at

9. messages

Stores individual messages within a conversation

- id
- conversation_id
- sender_id
- body
- is_read
- created_at

10. trade_requests

Handles exchange of the items between users

- id
- requester_id
- requested_item_id
- offered_item_id
- status
- created_at
- updated_at

11. reports

Stores reports made by users about inappropriate items or accounts.

- id
- reporter_id
- reported_user_id
- reported_item_id
- reason
- status
- created_at

Media storage: Uses file system. Images uploaded by users are stored in a dedicated uploads directory on the application server. The database stores only the relative file path to the image.

Search / Filter Implementation: Meilisearch is used for searching and filtering. It runs in a dedicated container and hooks into the sql container for the database and can be called in the app container.

Added frameworks / tools: Meilisearch, mentioned previously, and Multer. Multer is a node component to handle uploads.

6. Identify actual key risks for your project at this time

Risk - Skills: The team may have varying levels of experience with the chosen tech stack, particularly around the database design and building a fully functional marketplace.

Resolution: Identify the skill gaps early by having members self-report comfort levels. Pair stronger members with those who need support, and use resources like tutorials or office hours.

Risk - Scheduling: With team members balancing coursework, jobs, and/or other commitments, maintaining a consistent 2-3 day meeting schedule and hitting deadlines may be difficult.

Resolution: Use Trello to set internal deadlines ahead of actual due dates. Flag delays early in Discord so tasks can be redistributed if needed.

Risk - Technical: Some technical unknowns can include how to restrict marketplace access to verified SFSU students and how to structure the database to handle listings, users, and transactions efficiently.

Resolution: Research authentication methods (i.e. SFSU email domain) and review similar open-source marketplace projects early.

Risk - Teamwork: Uneven contributions across team members, especially as workload increases.

Resolution: Use Trello card assignments to track who is responsible for what, and address participation issues directly and early through the team lead.

Risk - Legal/Content: Any third party libraries, frameworks, or assets used in the marketplace must be properly licensed for use in this student project.

Resolution: Stick to open-source tools with permissive licenses and document all dependencies. Avoid using unlicensed images or content in any part of the project.

7. Project management:

To manage Milestone 2, our team utilized Trello to organize and track all required tasks, including the UI storyboard mockups, database organization, and functional requirements documentation. We assigned front-end and back-end help, in addition to the leads, to help with the UI mockups and database organization, with detailed card descriptions and individual assignments to ensure clear ownership and accountability. As team lead, I scheduled regular meetings to review progress on the Milestone 2 document, conduct check-ins with both the front-end and back-end teams, and address any outstanding questions or expectations for the milestone. On average, the team met up every 2-3 days, accommodating members' varying schedules when necessary. To verify that we were available, we used when2meet. We used Discord to communicate and/or reach out to those who needed to check in independently within their front and back-end peers and to facilitate group updates. We will continue to use these tools and methods for future tasks in the project.

8. Use of GenAI tools like ChatGPT and copilot (mandatory) –focus only on usage in Milestone 2:

Tool: ChatGPT, GPT-5.2, Web Interface

Usage:

It was used to assist in implementing image upload functionality using the multer middleware in Node.js. It helped configure disk storage , file size limits and validation.

It was also used to generate SQL scripts to populate database tables with structured sample data in order to test search and filtering functionality.

Usefulness : HIGH

Prompt:

“Write a multer configuration for Node.js that uploads images to a public/uploads folder, limits file size to 5MB.”

“Generate SQL script to insert sample items in the tables for testing search filters.”

Benefit: It accelerated backend development, reduced syntax errors , and helped structure database test data efficiently. All generated code was reviewed, tested, and modified as necessary before integration.

9. Team Lead Checklist:

- So far all team members are fully engaged and attending team sessions when required **ON TRACK**
- Team ready and able to use the chosen back and front end frameworks and those who need to learn are working on learning and practicing **ON TRACK**
- Team reviewed suggested resources before drafting Milestone 2 **DONE**
- Team lead checked Milestone 2 document for quality, completeness, formatting and compliance with instructions before the submission **ON TRACK**
- Team lead ensured that all team members read the final Milestone 2 document and agree/understand it before submission **DONE**
- Team shared and discussed experience with GenAI tools among themselves **DONE**